Mobile Internet Growth and Services Trade *

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Abstract

This article studies how internet connectivity, through overall internet use and mobile network coverage, affects international trade, with a focus on services. Using data for over 100 countries from 2004 to 2019, we estimate a two-stage structural gravity model linking digital infrastructure to bilateral trade flows. A 10 percent rise in internet use in the importing country raises bilateral services trade by approximately 7 percent, with comparable effects on the exporter side. The strongest responses occur in telecommunications and information, financial, insurance, and education services, while the effects on goods trade are generally smaller. Mobile network coverage yields a more mixed picture: 2G-and-higher coverage is positively associated with trade in most broad sectors, while 3G-and-higher coverage has no detectable effect on aggregate trade but positive effects in services, concentrated in digitally deliverable industries. These results suggest that expanding internet access lowers trade costs and broadens participation in global services markets.

Keywords: Internet connectivity, Mobile broadband, Services trade, Structural gravity.
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*All errors are the authors'.

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1 Introduction

Since the early 2000s, the expansion of mobile broadband internet has successfully brought high-speed connectivity to many regions that previously lacked reliable fixed-line service. Even where fixed broadband was already available, the expansion of mobile devices brought unprecedented levels of convenience. Smartphones transformed how consumers discover, purchase, and consume services, and how firms coordinate and deliver them across borders. In many developing countries, mobile broadband has provided a cost-effective substitute for fixed-line networks, including through household devices such as mobile routers, dongles, or SIM-enabled home modems that use mobile networks as their primary source of connectivity. As a result, an increasing share of trade, particularly in services, is now mediated by digital communication tools available on mobile devices.

Unsurprisingly, trade in services has grown far faster than trade in goods, accounting for a rising share of global value added and employment. According to the United Nations Conference on Trade and Development, global service exports have grown at a rate of 9% year-over-year, approximately three times the rate of growth for trade in goods (Trade and Development 2025). Yet, significant frictions remain in cross-border service exchange, especially in sectors that depend on real-time communications or regulatory clearance. Hence, anything that reduces these frictions (such as mobile broadband) will presumptively increase trade volumes.

Services trade also deserves separate study because it behaves differently from goods trade. During the 2008–2009 trade collapse, services trade proved far more resilient than goods trade (Ariu 2016). At the same time, the two are closely intertwined: services such as transport, finance, and communications are inputs into goods trade, and a growing literature studies how the two types of trade are linked and how shocks are transmitted between them (Nordås 2010; Eaton and Kortum 2018). Services are also the part of trade in which communication technology should matter most directly, since many services can now be delivered entirely over a network. Focusing on services does, however, raise a measurement challenge. Services are supplied internationally through four modes—cross-border supply, consumption abroad, commercial presence, and the movement of natural persons—and available bilateral statistics do not capture all four. We return to this issue in Section 3, where we explain which modes our data cover and why those are the modes most likely to respond to internet access.

Existing research confirms that internet use facilitates trade, but much of the earlier literature, such as Freund and Weinhold (2002, 2004) and Choi (2010), focused on web hosts or fixed-line access. More recent studies focused on broadband growth (Nath and Liu 2017) and mobile phones (Rodriguez-Crespo, Marco, and Billon 2021) specifically. However, these studies do not distinguish between different generations of mobile telephony (2G vs. 3G vs. 4G, etc.)[1] nor incorporate the latest in structural gravity techniques that are now standard in the international trade literature.

We address four questions, each motivated by a gap in this literature. First, does internet access—and mobile network coverage in particular—increase bilateral trade in services? Earlier work finds positive associations between ICT and services trade using web hosts, internet users, or broadband subscriptions (Freund and Weinhold 2004; Choi 2010; Nath and Liu 2017), and recent studies find that digital connectivity lowers trade costs, especially for digitally deliverable services (Bellucci, Rubínová, and Piermartini 2025; Park and Choi 2025). Few of these studies use mobile network coverage, and fewer still estimate effects

1. 2G mobile networks allow for digital mobile voice and texting. 3G mobile networks allow for mobile email and web browsing. 4G mobile networks allow for HD video streaming. 5G mobile networks, which are not covered in this study, allow for ultra-low latency and device connectivity.

on services within a structural gravity framework. Second, do higher-capacity mobile generations matter more than basic coverage? Abeliansky and Hilbert (2017) and Abeliansky, Barbero, and Rodriguez-Crespo (2021) show that the quality of connections (bandwidth) matters for trade in addition to the number of subscriptions, especially in developing countries, and Hjort and Poulsen (2019) show that the arrival of fast internet had large labor-market effects in Africa. To our knowledge, no study compares mobile network generations in a gravity setting. We compare 2G-and-higher with 3G-and-higher coverage; data on 4G coverage are too sparse over our sample period to support estimation. Third, are the effects stronger on the importer side or the exporter side? Much of the early literature emphasizes exporters (Freund and Weinhold 2004; Clarke and Wallsten 2006), whereas Akerman, Leuven, and Mogstad (2022) and Hjort and Tian (2025) stress demand-side channels such as buyer search and matching. Because our framework includes connectivity on both sides of each bilateral pair, we can compare the two directly. Fourth, do the effects vary across service industries with different degrees of digital deliverability? Services trade costs are known to differ widely across sectors (Anderson et al. 2018), and Nath and Liu (2017) find ICT effects in several service categories. We estimate effects separately for seventeen service industries.

The data that we bring to bear on these questions include a dataset from the International Telecommunication Union that provides the share of individuals using the internet and population coverage of 2G-and-higher and 3G-and-higher mobile networks. We use bilateral trade flows from the International Trade and Production Database for Estimation (ITPD-E), which covers goods and services disaggregated across industries, together with bilateral gravity variables from the companion Dynamic Gravity Dataset, both published by the U.S. International Trade Commission. The granularity of these data allows us to observe both the extensive margin of connectivity (network rollout) and its intensive margin (network upgrading).

Our analysis shows that greater internet access significantly increases bilateral trade in services. A 10 percent rise in the share of internet users in the importing country is associated with about a 6.9 percent increase in services imports, while an equivalent rise in exporter connectivity raises services exports by roughly 6.8 percent. Among service industries, the elasticities are large and precisely estimated in information-intensive industries such as Telecommunications, Computer, and Information Services and Financial Services, where they range from about 0.80 to 0.87, and they are positive though smaller for Intellectual Property services at around 0.42. Importer elasticities are also large in Education and Travel, highlighting the role of consumer-side access in expanding cross-border service consumption. By contrast, the estimated effects for agriculture and manufacturing are smaller, between about 0.1 and 0.5, while mining shows a large importer elasticity (0.72) but no exporter effect. Results using mobile network coverage are more mixed. A 10 percent increase in 2G-plus coverage is associated with increases in aggregate trade of approximately 3.0 percent on the exporter side and 5.1 percent on the importer side. 3G-plus coverage has no detectable effect on aggregate trade, but it yields positive elasticities of about 0.3 to 0.4 in services, significant at conventional levels in telecommunications and information services, education, construction, and other business services, and marginally significant in finance and insurance. These results indicate that mobile connectivity broadly reinforces the pattern found for overall internet access, with the clearest effects in service industries where information transmission and coordination are central to trade.

This study advances the literature on ICT and services trade along five fronts. First, we construct a cross-country panel of internet use and mobile network coverage. We assemble annual country-level measures from the ITU, namely the share of individuals using the internet and the population coverage of 2G-and-higher and 3G-and-higher mobile networks, and link them to bilateral trade flows for broad sectors and detailed

service industries. This dataset allows us to distinguish between network rollout—the extensive margin of internet access—and network upgrades, the intensive margin, which capture improvements in bandwidth and quality.

Second, we embed internet access into a modern structural gravity framework using a two-step identification strategy. In the first stage, we estimate bilateral trade flows using PPML with exporter–time and importer–time fixed effects, which absorb the multilateral resistance terms and address zero trade flows and heteroskedasticity. Because country-level internet measures are perfectly collinear with these fixed effects, they cannot be identified in the first stage. In the second stage, we therefore hold the bilateral trade-cost estimates from the first stage fixed and replace the exporter–time and importer–time fixed effects with the internet measures together with exporter, importer, and year fixed effects. This approach provides a transparent and replicable framework for identifying the role of digital infrastructure in shaping international trade patterns.

Third, we document elasticities, asymmetries, and sectoral heterogeneity in the trade effects of internet access. Importer-side effects are at least as large as exporter-side effects in most specifications, and clearly larger in aggregate trade, agriculture, mining, and consumer-facing services such as travel and education; for total services, the two are nearly identical. Digitally deliverable services—such as telecommunications and IT, finance, and insurance—exhibit large elasticities, while intellectual property services show smaller ones. Goods trade provides a useful point of comparison: the estimated effects are smaller for agriculture and manufacturing than for services. We also compare results across alternative functional forms, including coverage levels versus logarithmic specifications, and contrast 2G-plus and 3G-plus coverage to separate the effects of basic connectivity from those of higher-capacity networks.

Fourth, our findings yield policy-relevant implications for digital trade and economic development. By identifying where internet access matters most, and on which side of the border its effects are strongest, the analysis informs the design of digital provisions in trade agreements, the prioritization of universal service obligations, and the targeting of development assistance aimed at strengthening services export capacity and improving market access.

Finally, the study contributes to a deeper understanding of international services trade. Empirical research on services remains limited relative to goods, largely because services are less standardized, often require regulatory approval or physical presence, and are more difficult to measure consistently across countries. Existing work emphasizes that services differ from goods not only in their production and delivery modes but also in their sensitivity to information frictions, coordination costs, and policy barriers. We provide new quantitative evidence on how digital infrastructure reduces these frictions and reshapes the geography of trade in information-intensive and consumer-facing services.

The remainder of the paper is organized as follows. Section 2 reviews related work on internet connectivity, mobile broadband, services trade, and gravity estimation. Section 3 describes the data sources and the construction and interpretation of the key variables, including internet usage, mobile coverage, and bilateral trade flows. Section 4 presents the structural gravity framework that motivates the empirical specification, and Section 5 outlines the estimation strategy. Section 6 reports the empirical results by baseline specification, sectoral heterogeneity, and mobile network generations. Section 7 provides interpretation and broader implications. Section 8 concludes.

2 Literature Review

2.1 Internet and Trade

Two of the first papers to look at the effect of the role of ICT infrastructure, especially Internet penetration, in reducing trade costs in services are Freund and Weinhold (2002, 2004). Both find that the growth of the Internet has had a significant positive effect on trade in services. The 2004 paper uses time-series and cross-sectional data to estimate that a 10% increase in web hosts in a country is associated with a 0.2% - 0.4% increase in services trade volume. Vemuri and Siddiqi (2009), analyzing panel data for 64 countries (1985–2005), likewise concluded that ICT infrastructure and internet availability have a positive, significant impact on trade volumes. Notably, they found this effect for total trade in goods and services, highlighting ICT as a facilitator of globalization broadly. Similarly, Choi (2010) uses a gravity model with a fixed-effects GMM estimator to find that a doubling of Internet usage in a country leads to an increase in the trade of services of 2% - 4%. Liu and Nath (2013) focus on 40 emerging countries from 1995-2010 across multiple measures of ICT and find that greater ICT use – particularly with respect to internet subscriptions and hosts – is correlated with higher export and import shares in those economies. Nath and Liu 2017 look specifically at broadband penetration in developing countries and find that it is correlated with increased trade in several service categories. Furthermore, this effect seems to be more pronounced in developing economies than in developed ones. Finally, Kang (2020) looks at the expansion of mobile broadband specifically in South Korea and finds that the effect of mobile Internet penetration may have had an even greater impact on the trade in services than fixed line broadband.

While the literature on the relationship between ICT infrastructure and services trade is still developing, there is also an extensive literature looking at the relationship between ICT and goods trade. Fink, Mattoo, and Neagu (2005) look at the elasticity between communications costs and trade, finding that a 10% reduction in communications costs leads to an 8% increase in trade. Tang (2006) applies the same analysis to the variety of goods traded and finds that the post 1970s reduction in communication costs from improvements in ICT in the United States led to a significant increase in the variety of goods imported into the U.S. Clarke and Wallsten (2006) demonstrate that internet access improvements in low-income countries translate into increased exports to rich markets. This implies ICT helps developing-country firms overcome traditional barriers (like poor information or high communication costs) that previously limited their export reach. At the firm level, Kneller and Timmis (2016) provided micro-level evidence from the UK, showing that the availability of high-speed broadband internet can increase the extensive margin of service exports—i.e., more firms start exporting services. Switching to Europe and the impact of ICT on exports, Mattes, Meinen, and Pavel (2012) find that the countries in the European Union with higher levels of ICT adoption have greater export intensity.

A more recent literature focuses more specifically on mobile broadband and trade. An early paper in this literature, Chung, Fleming, and Fleming (2013), studied APEC countries' agricultural trade: they found internet use helped trade, but the largest effect was from fixed-line phones, with no significant effect from mobile phones. They speculate that either data limitations or the timeframe (1997–2006, when mobile internet was nascent) might explain why mobile effects were not yet evident. However, in later literature the consensus is that mobile broadband connectivity has become a critical factor for trade, especially in regions where cellular networks are the main conduit for business and consumer internet access. Ahmad, Ismail, and Hook (2011), who study Malaysia specifically, find that mobile and fixed telephone subscriptions

have the biggest impact on trade flows, more than just internet users. Xing (2018) analyzes ICT and e-commerce adoption's impact on bilateral trade flows across 51 countries. It finds that ICT significantly increases exports, particularly for developing countries integrating into global value chains. Rodriguez-Crespo, Marco, and Billon (2021) look at the relative impacts of the Internet, mobile phones, and broadband penetration on trade flows and conclude that mobile phones have a greater short-run impact on trade than fixed location Internet, including broadband. Abeliansky and Hilbert (2017), using data on both the number of subscriptions and the bandwidth of connections, find that both affect exports, with the number of subscriptions mattering most for developed countries and connection speed mattering most for developing countries. Abeliansky, Barbero, and Rodriguez-Crespo (2021) similarly find that both the quantity and the quality of ICT connections matter for the margins of trade, with quality especially important for developing countries. Mulenga and Mayondi (2022) look more at the impact of growth in digital services trade on GDP and find that it has a positive impact, especially in developing countries. However, in the process, the paper observes that ICT infrastructure is an important determinant of growth in that trade.

Several recent studies sharpen these findings. Exploiting the staggered rollout of broadband in Norway, Akerman, Leuven, and Mogstad (2022) show that broadband adoption makes firms' trade more responsive to distance and to partners' economic size, consistent with internet access lowering information frictions on the buyer side. Islam et al. (2024) estimate a PPML gravity model for 130 countries and find that ICT use in both exporting and importing countries raises merchandise exports, with effects that depend on education human capital. For sub-Saharan Africa, Billon, Rodríguez-Andrés, and Rodríguez-Crespo (2023) find that broadband use raises bilateral trade and mitigates the trade-reducing effects of long border-clearance times, and Bello et al. (2025) find that ICT diffusion promotes inclusive growth and complements trade liberalisation. Park and Choi (2025) find that ICT development in G20 countries is associated with higher trade in both goods and services, with diminishing returns as ICT development advances. Most closely related to our work, Bellucci, Rubínová, and Piermartini (2025) find that a 10 percentage point increase in digital connectivity lowers bilateral trade costs by about 2 percent on average, with larger effects for digitally deliverable services and in economies with open digital regulation. Finally, Hjort and Tian (2025) review the evidence on internet connectivity in developing countries and conclude that effects vary considerably with the setting, the outcome, and the type of internet access.

Hence, the literature points to an important effect of the growth in ICT infrastructure on growth in services trade, and mobile broadband may have the greatest impact of all. Our study contributes to this literature by constructing a cross-country panel of internet use and mobile network coverage by generation, embedding it in a structural gravity model, and estimating its effects separately for goods sectors and detailed service industries.

2.2 Trade in Services

Compared with goods, there is relatively limited empirical research on international trade in services, largely because of persistent data and measurement challenges. Services are less standardized, often require direct interaction between producers and consumers, and are subject to a wide range of domestic regulations that restrict cross-border supply. As a result, official trade statistics understate the true scale of services trade. Nevertheless, existing studies highlight that services account for most global value added and employment, and that openness in services markets can significantly enhance productivity in both services and manufacturing. Producer services such as telecommunications, finance, and logistics are central to enabling

transactions and coordination across borders, reinforcing their role as a foundation for modern trade and global value chains.

Services trade also responds to shocks differently from goods trade. Ariu (2016) shows that services trade fell far less than goods trade during the 2008–2009 collapse, mainly because services exports are less sensitive to income shocks in destination countries than goods exports, which are concentrated in durables. Goods and services trade are nonetheless closely linked, and Nordås (2010) and Eaton and Kortum (2018) study the connections between the two, including how shocks in one are transmitted to the other. These links imply that changes in the cost of trading services can also affect goods trade, which is one reason we report results for both.

Conceptual and quantitative work has shown that barriers in services trade are structurally different from those in goods markets. Francois and Hoekman (2010) emphasize that services are exchanged through multiple modes of supply, including cross-border delivery, consumption abroad, commercial presence, and movement of natural persons. Since explicit tariffs are rare, the main constraints are regulatory—licensing, ownership limits, and administrative discretion—which vary widely across sectors and countries (see also Hoekman and Braga 1997). Computable general equilibrium analyses, such as those by Christen, Francois, and Hoekman (2013), show that liberalizing service sectors can generate large indirect gains through productivity spillovers, as more efficient intermediate services improve competitiveness in downstream industries. These studies also stress that the scarcity and inconsistency of data remain key barriers to empirical analysis.

Recent empirical work has begun to map the scale and consequences of services trade using improved data and methods. Anderson et al. (2018) document that despite falling policy barriers, services face higher trade costs than goods, driven by information frictions and regulation (see also Miroudot, Sauvage, and Shepherd 2013). Fenske et al. (2021) show that the growing expenditure share of services—rising from roughly 60 percent in 1970 to nearly 80 percent in 2015—has moderated the aggregate growth of global trade, since many services remain less tradable even as demand for them expands. Firm-level evidence from Breinlich, Soderbery, and Wright (2018) and Lassmann (2020) shows that firms engaging in services exports tend to be more productive, pay higher wages, and rely more on skilled labor. Together, these findings underscore that services are both a growing source of comparative advantage and a critical input to other traded activities.

Building on this literature, our study introduces new evidence on how digital infrastructure reduces the distinctive frictions that limit cross-border services exchange. By linking internet use and mobile network coverage to bilateral trade flows, we estimate how connectivity affects trade in information-intensive and consumer-facing services.

2.3 The Gravity Model and Trade

The "Gravity Model" of international trade developed from an observation by Tinbergen (1962) that the volume of trade between two countries could be predicted by the product of the two countries' GDPs divided by the distance between them. The name of the model came from the similarity between his formula and that of the Newtonian gravitational pull between two objects. The gravity equation quickly became a workhorse of empirical international trade.

For a long time, however, the theoretical foundations of this empirical tool received little attention. Anderson (1979) provided the first theoretical derivation, and James E Anderson and van Wincoop (2003) showed that bilateral trade depends not only on bilateral trade costs but also on each country's trade costs

with all of its partners, which they summarize as "multilateral resistance" terms. Omitting these terms biases estimates of trade-cost effects, a problem that Baldwin and Taglioni (2006) label the "gold medal mistake." Because multilateral resistance terms are not directly observable, the standard practice in panel data is to absorb them with exporter–time and importer–time fixed effects (Head and Mayer 2014; Yotov et al. 2016).

A second development concerns the estimator. Silva and Tenreyro (2006) show that estimating the log-linearized gravity equation by OLS is inconsistent in the presence of heteroskedasticity and discards zero trade flows, and they propose the Poisson pseudo-maximum likelihood (PPML) estimator. PPML has since become the standard estimator for structural gravity (Head and Mayer 2014; Y. Yotov 2022). Fally (2015) shows that PPML with exporter and importer fixed effects has a further advantage: the estimated fixed effects are exactly consistent with the structural multilateral resistance terms implied by the model. In this sense, "structural gravity" in current practice refers to the combination of a theory-consistent specification, PPML estimation, and exporter–time and importer–time fixed effects.

These fixed effects create a well-known difficulty for studying country-level variables such as internet access. Any variable that varies only by country and year, such as the share of internet users in the exporting country, is perfectly collinear with the exporter–time fixed effects, so its effect cannot be identified when the full set of fixed effects is included. The literature offers two ways around this problem. The first is a two-step procedure, as in Anderson and Yotov (2016): the gravity equation is first estimated with the full set of fixed effects, and the country-level variables are then introduced in a second stage in which the exporter–time and importer–time fixed effects are replaced by the variables of interest and less restrictive fixed effects. The second, developed by Heid, Larch, and Yotov (2021), uses domestic trade flows alongside international flows, which allows country-specific variables that affect only international trade to be identified in a single step even with exporter–time and importer–time fixed effects; Y. V. Yotov (2022) discusses the broader importance of domestic trade flows for gravity estimation. We follow the first approach. In line with the recommendations of Yotov et al. (2016) and Y. V. Yotov (2022), our estimation sample includes domestic as well as international trade flows, and our first stage includes an indicator for domestic trade. Section 5 describes the procedure in detail.

Many of the papers cited above (with the exception of Choi (2010), Lin (2015), and some of the more recent studies) do not fully engage with this literature. Hence, estimates of the effects of ICT on bilateral trade may not be consistent in these studies. This study contributes to the literature by fully engaging with the literature on structural gravity modeling and the impact of ICT on the trade in services.

3 Data

3.1 Data on Internet Use and Mobile Network Coverage

Our data on internet access come from the International Telecommunication Union (ITU).[2] We use annual country-level measures of the share of individuals using the internet and the population coverage of 2G-and-higher (2G+) and 3G-and-higher (3G+) mobile networks. Our main measure is the share of internet users. We complement it with 3G+ coverage, which supports web browsing and social media and expanded broadly across countries during our study period, and with 2G+ coverage, which includes basic 2G connectivity and

2. https://datahub.itu.int/data/

all higher generations and therefore captures basic network availability. To our knowledge, this is the most comprehensive source that jointly spans a long time horizon and broad country coverage, making it well suited for panel analysis of trade flows.

Ideally, we would also incorporate 4G-and-higher and 5G-and-higher coverage as additional bandwidth measures that should, in principle, capture greater reductions in service-sector trade costs. In practice, those series exhibit substantial missingness across countries and years, and even a conservative two-year interpolation leaves large gaps. We therefore focus on 2G+ and 3G+ coverage, which provide the best balance between temporal continuity and global coverage.

Figure 1 plots the share of internet users by country and year. Figures 2 and 3 plot 2G+ and 3G+ mobile coverage, respectively. Each panel is a heat map where the printed numbers represent percentages. These figures document the diffusion of connectivity over time: widespread 2G+ availability arrives earlier and saturates at high levels, while 3G+ expands later and displays richer within-country dynamics during our study window.[3]

3.2 What the Connectivity Measures Capture

The three measures capture different aspects of connectivity, and it is important to be precise about what each one measures. The share of internet users is a measure of *use*: the ITU defines it as the proportion of individuals who used the internet from any location in the last three months, regardless of the device or type of connection. It therefore combines fixed and mobile access. The 2G+ and 3G+ coverage measures, by contrast, are measures of *availability*: they record the share of the population living within range of a mobile signal of the given generation or higher, irrespective of whether those people subscribe to or use the service. Coverage can therefore exceed use substantially, especially in lower-income countries where devices and data plans are expensive relative to income.

Both types of measure share two limitations. First, they are bounded at 100 percent and approach saturation in many countries. This is especially true of 2G+ coverage, which reaches close to universal levels in most countries early in our sample, and of internet use in high-income countries by the end of the sample. Because our second-stage estimates include country fixed effects, they are identified from changes within countries over time, so the estimates for 2G+ in particular are driven by the subset of countries in which coverage was still expanding. Near the upper bound, small changes in coverage correspond to very small changes in logs, which helps explain why some of the 2G+ elasticities reported below are very large and imprecisely estimated. We report specifications in levels as well as logs for this reason.

Second, these measures capture the quantity of connectivity rather than its quality. As Abeliansky and Hilbert (2017) and Abeliansky, Barbero, and Rodriguez-Crespo (2021) emphasize, two countries with the same share of connected individuals can differ greatly in connection speed, reliability, and the amount of information that can be transmitted, and the quality of connections matters for trade in addition to the number of subscriptions, particularly in developing countries. Measures of information capacity, such as those proposed by Hilbert (2022), would allow a finer analysis of these differences, but they are not available for our panel of countries and years. The comparison between 2G+ and 3G+ coverage provides only a coarse proxy for quality, since 3G+ networks support substantially higher data transmission rates than 2G networks.

3. Appendix A reproduces these panels after applying a simple two-year interpolation to fill short gaps in the raw ITU series. The interpolated visuals clarify regional diffusion patterns where annual data are sparse. Importantly, no interpolation is used in any estimation or quantitative result.

Our results should therefore be read as the effects of the spread of connectivity rather than of improvements in its quality, and the comparison between mobile generations as indicative rather than definitive evidence on the role of bandwidth.

3.3 Data on International Trade

Our trade data come from the International Trade and Production Database for Estimation, Release 2 (ITPD-E-R02), published by the U.S. International Trade Commission (Borchert et al. 2022).[4] We use data for the period 2004–2019. The ITPD-E provides annual bilateral trade flows between countries, measured in U.S. dollars and disaggregated into 170 industries, of which industries 154 to 170 are services. Its country codes are synchronized with the USITC Dynamic Gravity Dataset (version 2), from which we take the bilateral gravity variables—distance, contiguity, common language, colonial ties, and free trade agreements.[5] Trade values reflect total exports from the origin country to the destination country. An important feature of the ITPD-E is that it reports domestic trade flows—each country's sales to itself—alongside international flows, constructed consistently from production and trade data. Following Y. V. Yotov (2022), we include these domestic flows in our estimation sample. Doing so ensures that the estimated effects of trade costs are measured relative to a country's trade with itself, which is the comparison implied by structural gravity theory, and avoids the bias that Y. V. Yotov (2022) shows can arise when domestic trade is omitted.

We focus on aggregate trade as well as trade in different sectors and industries, using the sectoral and industry identifiers provided in the dataset. For most of the analysis, we use total trade as the outcome, but we also examine patterns by broad industry group—agriculture, manufacturing, mining, and services—and by individual service industry. All trade flows are measured in current U.S. dollars and matched by country-pair and year to other data sources used in the analysis.

The services trade data in the ITPD-E are drawn from balance-of-payments statistics, mainly the WTO–UNCTAD–ITC annual trade in services database supplemented with UN trade in services data (Borchert et al. 2022). Balance-of-payments statistics record transactions between residents and nonresidents. They therefore capture cross-border supply (Mode 1) and consumption abroad (Mode 2) well, and the movement of natural persons (Mode 4) only partially, but they do not capture services supplied through commercial presence abroad (Mode 3), which is recorded in foreign affiliate statistics instead (Francois and Hoekman 2010). This limitation is less severe for our question than for many others. The channels through which internet access should affect services trade—remote delivery of services and the ability of consumers to search for, compare, and purchase services abroad—operate primarily through Modes 1 and 2, which our data do capture. Our estimates should nevertheless be interpreted as effects on services trade excluding commercial presence; they do not tell us whether internet access complements or substitutes for services supplied through foreign affiliates.

4 Theoretical foundations

This section briefly presents a standard structural gravity model. For simplicity in notation, we do not include product-level indicators. However, as demonstrated by Yotov et al. (2016), the inclusion of product-level indicators does not substantially change the structure of the model.

4. https://www.usitc.gov/data/gravity/itpde.htm
5. https://www.usitc.gov/data/gravity/dgd.htm

4.1 Structural Gravity

Consumer preferences are identical across countries, and consumers' utility U function for country j follows a CES structure:

$$U(c_j) = \left\{ \sum_i \alpha_i^{\frac{1-\sigma}{\sigma}} c_{ij}^{\frac{\sigma-1}{\sigma}} \right\}^{\frac{\sigma}{\sigma-1}} \tag{1}$$

where c_{ij} is consumption of varieties from country i in country j, $\sigma > 1$ is the elasticity of substitution between varieties of goods from different countries, and $\alpha_i > 0$ is the CES preference weight parameter, which can be interpreted as product quality or appeal.

Consumers maximize their utility function subject to the following budget constraint:

$$\sum_i p_{ij} c_{ij} = E_j, \tag{2}$$

where E_j is the total spending on varieties from all countries. The delivered price in country j of the good from country i, $p_{ij} = p_i t_{ij}$, is determined by the f.o.b. export price of the good p_i from country i and the bilateral "iceberg" trade costs t_{ij}.

Following James E Anderson and van Wincoop (2003, 2004) we can use the CES utility function and a general equilibrium setting to solve for a general equilibrium model of trade flows between countries. The gravity equation that governs bilateral trade between any two countries i and j, X_{ij}, is as follows:

$$X_{ij} = \frac{Y_i E_j}{Y} \left(\frac{t_{ij}}{\Pi_i P_j} \right)^{1-\sigma}, \tag{3}$$

where Y_i denotes the value of production (total sales) of country i, E_j denotes total expenditure of country j, and $Y = \sum_i Y_i$ is world output. The key insights from this literature are the "multilateral resistance terms" $\Pi_i^{1-\sigma}$ and $P_j^{1-\sigma}$. These terms link the partial equilibrium and the general equilibrium impacts of trade costs. Mathematically, they are CES price indexes. The "outward" multilateral resistance term $\Pi_i^{1-\sigma}$ summarizes the trade costs that exporter i faces with all of its destination markets and is equal to:

$$\Pi_i^{1-\sigma} = \sum_j \left(\frac{t_{ij}}{P_j} \right)^{1-\sigma} \frac{E_j}{Y}. \tag{4}$$

The "inward" multilateral resistance term $P_j^{1-\sigma}$ is the CES price index of imports to country j and is equal to:

$$P_j^{1-\sigma} = \sum_i \left(\frac{t_{ij}}{\Pi_i} \right)^{1-\sigma} \frac{Y_i}{Y}. \tag{5}$$

Taking equation (3) to panel data, we can collect all terms that vary only by exporter and year and all terms that vary only by importer and year:

$$X_{ijt} = \exp\left[\delta_{it} + \gamma_{jt} + (1-\sigma) \ln t_{ijt} \right], \tag{6}$$

where

$$\delta_{it} \equiv \ln Y_{it} - (1-\sigma) \ln \Pi_{it} - \ln Y_t, \qquad \gamma_{jt} \equiv \ln E_{jt} - (1-\sigma) \ln P_{jt}.$$

The exporter–time term δ_{it} thus contains the exporter's output and outward multilateral resistance, and

the importer–time term γ_{jt} contains the importer's expenditure and inward multilateral resistance. Because these terms are unobserved and depend on trade costs with all partners, they are absorbed by exporter–time and importer–time fixed effects in estimation.[6] As noted by Silva and Tenreyro (2006), a log-linear OLS specification of this equation is generally inappropriate for two reasons. The first is that many bilateral trade flows will have zero observations and will therefore be dropped from a log-linear sample in a way that is unlikely to be random. The second is that log-linear OLS may be biased if there is considerable heteroskedasticity in the sample. Consequently, the authors suggest that a Poisson pseudo-maximum likelihood (PPML) estimator will best address the econometric issues arising in a gravity specification, and PPML with exporter–time and importer–time fixed effects is the standard estimator of structural gravity (Head and Mayer 2014; Fally 2015; Y. Yotov 2022).

4.2 Trade Effects of Internet

We define the trade cost of a good $t_{ijt}^{1-\sigma}$ from country i to j in year t in the following functional form:

$$t_{ijt}^{1-\sigma} = \exp\left[\beta_1 \text{Internet}_{it} + \beta_2 \text{Internet}_{jt} + \boldsymbol{\beta}_3' \mathbf{z}_{ijt}\right]. \tag{7}$$

The variables Internet_{it} and Internet_{jt} represent the internet access measures of exporting country i and importing country j, respectively. The vector $\mathbf{z}_{ijt}$ contains bilateral trade-cost variables: distance between i and j and indicators for a free trade agreement, contiguous borders, common language, and colonial ties. Internet access enters as a country-specific, or non-discriminatory, trade cost: better connectivity in country i is assumed to lower the cost of trading with all of its partners.

Substituting equation (7) into equation (6) gives

$$X_{ijt} = \exp\left[\delta_{it} + \gamma_{jt} + \beta_1 \text{Internet}_{it} + \beta_2 \text{Internet}_{jt} + \boldsymbol{\beta}_3' \mathbf{z}_{ijt}\right]. \tag{8}$$

Internet access affects bilateral trade through two channels. The direct channel is the reduction in bilateral trade costs captured by β_1 and β_2. The indirect channel works through general equilibrium: lower trade costs change the multilateral resistance terms in equations (4) and (5), and connectivity may also raise output and expenditure, Y_{it} and E_{jt}, for example by improving productivity. Equation (8) also makes clear the identification problem discussed in Section 2: Internet_{it} varies only by exporter and year, exactly like δ_{it}, so the two cannot be separately identified when exporter–time fixed effects are included, and likewise for Internet_{jt} and γ_{jt}. The next section describes how we address this.

5 Empirical Analysis

We estimate equation (8) in two steps, following Anderson and Yotov (2016). Let X_{ijt}^s denote bilateral trade (in levels) from exporter i to importer j in year t and sector or industry s.

In the first step, we estimate the standard structural gravity equation by PPML:

$$X_{ijt}^s = \exp\left[\boldsymbol{\beta}_3' \mathbf{z}_{ijt} + \kappa D_{ij} + \lambda_{st} + \delta_{it} + \gamma_{jt}\right] \times \varepsilon_{ijt}^s, \tag{9}$$

6. James E Anderson and van Wincoop (2003) show that omitting multilateral resistance biases estimates of trade-cost effects; Baldwin and Taglioni (2006) call this omission the "gold medal mistake."

where the sample includes both international flows ($i \neq j$) and domestic flows ($i = j$), D_{ij} is an indicator equal to one for domestic flows, λ_{st} are sector–time fixed effects, and δ_{it} and γ_{jt} are exporter–time and importer–time fixed effects. The domestic-trade indicator captures the average difference between domestic and international trade that is not explained by the bilateral variables, so that the coefficients on the bilateral variables are not contaminated by the very large volume of domestic trade. The exporter–time and importer–time fixed effects absorb output, expenditure, and the multilateral resistance terms in equation (6), as well as any other country–year determinants of trade, including internet access. This first stage therefore yields estimates of the bilateral trade-cost parameters $\hat{\boldsymbol{\beta}}_3$ that are consistent with structural gravity.

In the second step, we hold the bilateral component of trade fixed at its first-stage estimate and replace the exporter–time and importer–time fixed effects with the internet measures and with exporter, importer, and year fixed effects:

$$X_{ijt}^{s} = \exp\left[\widehat{B}_{ijt}^{s} + \beta_1 \text{Internet}_{it} + \beta_2 \text{Internet}_{jt} + \mu_i + \chi_j + \tau_t\right] \times \eta_{ijt}^{s}, \qquad \widehat{B}_{ijt}^{s} \equiv \hat{\boldsymbol{\beta}}_3' \mathbf{z}_{ijt} + \hat{\lambda}_{st}, \tag{10}$$

where $\widehat{B}_{ijt}^{s}$ enters with its coefficient constrained to one, and μ_i, χ_j, and τ_t are exporter, importer, and year fixed effects. We again estimate by PPML, on the same sample as the first stage. Because the second stage includes exporter and importer fixed effects rather than exporter–time and importer–time fixed effects, β_1 and β_2 are identified from changes in connectivity within each country over time, relative to the common year effects. This is what resolves the collinearity problem described above.

This design has an implication for interpretation. The exporter–time and importer–time fixed effects in the first stage absorb not only multilateral resistance but also output and expenditure. In the second stage, these terms are captured only by the time-invariant country effects and the common year effects. The coefficients β_1 and β_2 should therefore be interpreted as the effect of connectivity on a country's overall propensity to export or import, conditional on time-invariant country characteristics and common shocks. This combines the direct trade-cost channel in equation (7) with any associated changes in output, expenditure, and multilateral resistance. In the main specification, the internet measures enter in logs, so β_1 and β_2 are elasticities; we also report a specification in levels.

6 Empirical Results

This section presents the empirical results from our two-stage estimation strategy. We begin by validating the gravity framework with a first-stage regression that establishes expected effects of distance, contiguity, language, and trade agreements. We then turn to the second-stage results, which quantify the role of overall internet usage on bilateral trade flows, with particular attention to the service sectors. Next, we examine heterogeneity across service industries to assess which sectors are the most responsive to internet connectivity. We then compare the effects of 2G+ and 3G+ mobile network coverage. Finally, we interpret these findings in light of mechanisms emphasized in the literature and draw broader implications for digital trade and development policy.

6.1 First-Stage: Gravity Estimates

Panel A of Table 1 reports the first-stage results from the structural gravity estimation. The coefficients are highly consistent with theoretical expectations and provide strong validation of the model's structure.

Distance enters negatively and significantly across all sectors, ranging from −0.346 in services to −1.058 in mining, implying that a 1% increase in bilateral distance reduces trade in services by approximately 0.35% and in mining by roughly 1.06%. These magnitudes are in line with distance elasticities typically reported in the gravity literature, suggesting that the data construction and estimation design are sound.

The standard trade-facilitating covariates behave as expected. Contiguity and common language significantly increase trade volumes, capturing the importance of shared borders and linguistic proximity for information flows and contract enforcement. Colonial ties are also positively associated with bilateral trade, particularly in mining and services, which likely reflects persistent institutional linkages and lower informational asymmetries across historically connected economies. The presence of a free trade agreement strongly and significantly raises trade across all sectors, consistent with the structural interpretation that policy-driven reductions in trade barriers directly lower bilateral trade costs t_{ij}. These patterns confirm that the first-stage regression effectively recovers the underlying geography and policy determinants of trade costs.

The first-stage coefficients for the alternative samples used in the 2G+ and 3G+ specifications are nearly identical in both sign and magnitude, confirming that the baseline gravity relationships are stable across subsamples and specifications. To conserve space, these results are not shown here but are available in Appendix A. Collectively, the first-stage estimates provide a solid foundation for the second-stage analysis, where we replace the exporter–time and importer–time fixed effects with measures of digital infrastructure and exporter, importer, and year fixed effects to identify how internet connectivity affects bilateral trade flows.

6.2 Second-Stage: Effects of Internet Infrastructure

Panel B of Table 1 presents the second-stage estimates linking internet access to bilateral trade. Using the log of internet users (combining both fixed and mobile access), we find robust positive effects on trade volumes, with particularly strong results for services. For services, the elasticity with respect to ln(Internet users) is 0.713 on the exporter side and 0.726 on the importer side, both highly significant. In manufacturing, the elasticities are 0.504 (exporter) and 0.488 (importer); in agriculture, 0.111 (exporter) and 0.320 (importer). For mining, the importer elasticity is large and significant at 0.718, while the exporter elasticity (0.104) is small and statistically insignificant. At the aggregate level, the exporter and importer elasticities of 0.305 and 0.453, respectively, suggest that both sides' connectivity increases trade volumes, though the demand side plays a somewhat larger role.

These results fit naturally within the structural gravity framework, where the bilateral trade-cost function t_{ijt} depends multiplicatively on exporter and importer factors. Internet access reduces trade costs by lowering both fixed and variable costs of coordination, enhancing search and matching, and enabling the transmission of complex, information-intensive products and services. The importer-side elasticities are larger than the exporter-side elasticities for aggregate trade, agriculture, and mining, suggesting that demand-side frictions matter: when buyers and consumers in destination markets are more connected, it becomes easier to search for, contract with, and consume foreign products and services. For manufacturing and for services as a whole, however, the exporter and importer elasticities are nearly identical, indicating that connectivity on both sides of the transaction matters about equally for these sectors.

Comparing Internet Users, 2G+, and 3G+. Tables 2 and 3 extend the analysis to mobile network coverage, distinguishing basic network availability from the availability of higher-capacity networks. For 2G+

coverage, the aggregate elasticities are 0.314 (exporter) and 0.535 (importer), with both being statistically significant. At the sectoral level, the manufacturing sector shows sizable elasticities of 0.522 and 0.617, while mining records a very large importer elasticity of 1.117; the agriculture coefficients are small and insignificant. The services sector displays extremely large coefficients—about 3.78 on both sides with wide standard errors—which likely reflect the saturation of 2G+ coverage discussed in Section 3.2: because coverage is close to its upper bound in most countries, the remaining within-country variation is small, and the estimates are driven by a limited set of countries in which coverage was still expanding. These results suggest that basic mobile connectivity—the extensive margin of access—facilitated cross-border transactions in countries with limited fixed-line infrastructure, but the magnitudes for services should be interpreted with caution.

The 3G+ estimates are more moderate. The aggregate elasticities are close to zero and not statistically significant, and the coefficients for agriculture and manufacturing are also insignificant. Mining exhibits a significant importer elasticity of 0.179, while services show positive elasticities of 0.318 (exporter) and 0.324 (importer), each significant at the 10 percent level. Thus, 3G+ coverage is associated with higher trade in services but not with higher trade in goods. The 3G+ elasticities for services are smaller than those for internet users, which is unsurprising: the share of internet users is a measure of actual use that combines fixed and mobile access, whereas 3G+ coverage measures only the availability of one type of network.

Comparing the two mobile generations is less straightforward than it might appear. Because 2G+ coverage includes all areas covered by 3G and later networks, the two measures are nested, and they are estimated on different samples over periods in which they were at different stages of diffusion. The results therefore do not allow us to say that 3G networks had larger effects than 2G networks. What they do show is that the effects of 3G+ coverage are concentrated in services, and, as shown below, in the service industries in which digital delivery is most feasible. This pattern is consistent with the view that higher-capacity networks matter most for services that depend on richer data exchange, such as finance and IT-enabled services, but given the limitations of coverage as a quality measure, we regard it as suggestive. Taken together, the results are consistent with the internet's contribution to trade operating both through extensive-margin participation, driven by access, and through intensive-margin efficiency, driven by capacity, although our country-level data do not allow us to decompose trade flows along these margins directly.

6.3 Heterogeneity by Service Industry

Table 4 reports substantial heterogeneity across service industries. The effects of internet usage are large and precisely estimated for digitally deliverable and information-intensive services, where transmission, coordination, and data handling are central. In Telecommunications, Computer, and Information Services (industry 162), the exporter and importer elasticities are 0.856 and 0.824, both highly significant. Financial Services (industry 160) display elasticities of 0.802 (exporter) and 0.865 (importer), and Insurance and Pension Services (industry 159) display elasticities of 0.814 and 0.809. Intellectual Property (industry 161) records smaller elasticities of 0.420 and 0.414. These coefficients align with the mechanism in which the internet reduces trade costs by enabling digital delivery, synchronized communication, and real-time verification between trading partners. Large effects are not confined to information-intensive services, however: Construction (industry 158) shows elasticities of about 1.02 on both sides, and Maintenance and Repair (industry 155) shows a large exporter elasticity of 1.057, which may reflect the role of connectivity in coordinating project-based work and remote diagnostics.

Consumer-oriented sectors show stronger importer-side elasticities, emphasizing the demand-side role of connectivity. For Travel (industry 157), elasticities are 0.424 (exporter) and 0.762 (importer), indicating that digitally connected consumers are better able to search, compare, and purchase foreign services. In Education Services (industry 166), the elasticities are 0.809 (exporter) and 0.959 (importer), both significant, highlighting the growing global market for online and hybrid learning where student-side access is essential. These importer-dominant patterns are consistent with a gravity interpretation in which digital access lowers buyers' search and transaction costs for foreign services.

By contrast, sectors with inherently local delivery or administrative orientation show weaker or more mixed effects. Government services (industry 167) display no exporter responsiveness (-0.020) but a significant importer elasticity of 0.572, which more likely reflects institutional or programmatic cross-border transactions than market-driven trade. Heritage and Recreational Services (industry 164) exhibit small and insignificant coefficients (0.170 exporter, 0.046 importer), consistent with physical presence requirements and local consumption. Such results reinforce the idea that digital infrastructure primarily facilitates tradable services whose delivery can be decoupled from geography.

Tables 6 and 5 report the corresponding results for 3G+ and 2G+ coverage. Under 2G+, the estimates are large but imprecise and mixed in sign. In industry 162, exporter and importer elasticities reach 3.823 and 3.553, while Financial Services (160) record 3.709 (exporter, significant) and 2.753 (importer, insignificant); several other industries, including Intellectual Property (161) and Heritage and Recreation (164), have negative but insignificant coefficients. As with the sectoral results, these magnitudes likely reflect limited within-country variation in 2G+ coverage near its upper bound, and we do not place much weight on them. Where the estimates are positive and significant, they are consistent with the view that even low-bandwidth connectivity expanded the extensive margin of participation in international service markets, especially for countries lacking robust fixed-line infrastructure.

Under 3G+, the elasticities are more moderate and more informative. In industry 162, exporter and importer elasticities are 0.356 and 0.395, both significant at the 5 percent level, and Financial Services (160) shows 0.342 on each side, significant at the 10 percent level. Intellectual Property (161) exhibits a significant importer elasticity of 0.103, while Education Services (166) registers 0.385 (exporter) and 0.282 (importer), both significant. Construction (158) and Other Business Services (163) also show significant elasticities of about 0.30 to 0.46. These results suggest that the availability of higher-capacity mobile networks is associated with more trade in services that rely on data transmission and remote interaction, consistent with bandwidth upgrades operating on the intensive margin.

Heritage and Recreation (164) remains unresponsive to 3G+ coverage, with negative and insignificant coefficients. Government services (167) show a small but significant exporter elasticity (0.158) and no importer effect, which, as with internet users, is difficult to interpret as market-driven trade. Overall, the evidence across internet users, 2G+, and 3G+ indicates that connectivity is associated with more trade in services, and that the effects are largest where technology can most directly substitute for physical proximity.

7 Interpretation and Broader Implications

The estimates are best interpreted within a structural gravity framework in which connectivity variables alter bilateral trade costs and, through general equilibrium, shift multilateral resistance on both exporter and importer sides. Mobile internet affects several margins simultaneously. Rollout increases the number of

connected agents, reducing fixed costs of search, contracting, authentication, and coordination. Bandwidth upgrades improve reliability and speed, lowering variable costs for data-intensive and synchronous exchanges. The two-step estimation design, which first absorbs country–time fundamentals and then replaces them with measured connectivity and country and year fixed effects, follows the standard gravity structure described by James E Anderson and van Wincoop (2003) and extended by Yotov et al. (2016). As discussed in Section 5, the second-stage coefficients capture the overall association between connectivity and a country's trade, including but not limited to the direct trade-cost channel.

Industry patterns align with established distinctions in services tradability and regulation. Information-intensive services such as telecommunications, computer and information, finance, and insurance are either digitally delivered or rely on dense remote coordination. Reductions in information and coordination costs therefore yield larger elasticities in these sectors. By contrast, proximity-bound sectors such as heritage and recreation respond weakly, consistent with their limited scope for technology-based substitution. Structural gravity work on services by Anderson et al. (2018) documents large but uneven declines in border barriers and ties these differences to geography, income, infrastructure, and institutions. Similarly, Francois and Hoekman (2010) emphasize that cross-border services are strongly conditioned by behind-the-border rules governing entry, data, and professional licensing. These results are consistent with the heterogeneity observed across service categories in the estimated elasticities.

Importer connectivity typically matters at least as much as exporter connectivity. Buyer-side access reduces search and verification frictions, expands the potential customer base, and improves the matching efficiency for international transactions. Hjort and Tian (2025) highlight precisely these demand-side channels in addition to supply-side productivity gains, and Akerman, Leuven, and Mogstad (2022) provide direct evidence that broadband lowers buyers' information frictions. When consumers and firms in destination markets are connected, they can discover and transact with foreign providers more easily, which raises trade flows even if exporter-side infrastructure remains constant. This interpretation is consistent with the larger importer elasticities estimated for aggregate trade and for consumer-facing services such as travel and education.

The distinction between the extensive margin of connectivity (network rollout) and its intensive margin (bandwidth upgrades) is also central. Early mobile networks primarily expanded coverage and enabled participation, while later generations enhanced data throughput and reduced latency. The evidence reviewed by Hjort and Tian (2025) indicates that the gains from connectivity depend on the type of internet access, and Abeliansky, Barbero, and Rodriguez-Crespo (2021) find that connection quality matters for trade in addition to the number of subscriptions. The concentration of the 3G+ effects in digitally intensive services is consistent with this channel, although, as noted above, coverage measures provide only a coarse proxy for connection quality.

Firm-level evidence from micro studies supports complementary supply-side adjustments. Breinlich, Soderbery, and Wright (2018) show that UK firms shifted from selling goods to selling services as tariffs on manufacturing inputs declined, with greater responses among firms with higher R&D intensity. This pattern of servicification suggests that improved connectivity and lower trade barriers can induce firms to expand into services exports, particularly in producer and knowledge-based activities. The same logic may help explain the large exporter-side elasticities for finance, insurance, and information technology services.

At the macro level, structural change provides context for interpreting magnitudes. Services now account for about two thirds of global value added and roughly half of employment. The share of services in final

expenditures rose from about 58 percent in 1970 to nearly 80 percent in 2015, while the ratio of services trade to services expenditure increased from roughly 8 to 18 percent (Fenske et al. 2021). Counterfactual exercises suggest that if global expenditure shares had remained at 1970 levels, openness in 2015 would have been about one third higher despite falling trade costs. These patterns imply that strong service-sector elasticities can coexist with a slower aggregate trade-to-GDP ratio because demand has shifted toward sectors with inherently lower tradability.

Policy and regulatory context determine how much connectivity translates into actual cross-border flows. Quantitative decompositions of services trade costs reveal that despite widespread declines, barriers remain large and vary widely across sectors and partners (Anderson et al. 2018). Bellucci, Rubínová, and Piermartini (2025) find that the trade-cost reductions from digital connectivity are larger in economies whose digital regulations keep communications infrastructure open and allow cross-border data flows. Computable general equilibrium analyses of market access, such as Christen, Francois, and Hoekman (2013), show that relaxing regulatory constraints, enhancing transparency, and improving competition can deliver large welfare and trade gains, especially when paired with infrastructure improvements. The literature consistently finds that similar physical networks yield different outcomes depending on the regulatory quality and the efficiency of domestic services markets (Francois and Hoekman 2010).

Taken together, the evidence implies four main conclusions. First, internet access is associated with substantially more bilateral trade, especially in services, in a manner consistent with structural gravity and with observed declines in sector-specific barriers. Second, importer-side connectivity matters at least as much as exporter-side connectivity, and more in aggregate trade and in consumer-facing services, consistent with important demand-side frictions in services trade. Third, the effects of higher-capacity mobile networks are concentrated in data- and information-intensive service industries. Fourth, digital infrastructure alone is not sufficient: complementary reforms in regulation, competition, and services market access are likely to determine how effectively connectivity increases cross-border trade. These results align with established evidence on the multiple modes of service delivery, the importance of information and coordination costs, and the persistent heterogeneity in border barriers across sectors and partners.

8 Conclusion

This study provides new evidence on how internet connectivity, encompassing both overall internet use and mobile network coverage, shapes international trade patterns with particular emphasis on services. Using a dataset that links internet usage and mobile network coverage to bilateral trade flows within a structural gravity framework, the analysis shows that improvements in digital connectivity are associated with significantly more cross-border trade. The estimated elasticities indicate that both exporter and importer connectivity enhance services trade, with nearly identical effects on the two sides for services as a whole. A 10 percent rise in the share of internet users in the importing country increases bilateral services trade by roughly 7 percent, with a similar effect on the exporter side. These magnitudes suggest that digital access meaningfully lowers informational and coordination frictions in international service markets. By contrast, the effects are smaller for agriculture and manufacturing, while mining shows a large importer-side effect but no exporter-side effect.

The results further show that the impact of internet access varies substantially across service industries. Large effects appear in digitally intensive industries: Telecommunications, Computer, and Information Ser-

vices exhibit exporter and importer elasticities near 0.85, Financial Services show effects around 0.80 to 0.87, and Insurance services show effects around 0.81. Intellectual Property services record smaller but still positive coefficients near 0.42, and importer elasticities in Education and Travel services range from about 0.76 to 0.96. By contrast, Heritage and Recreational services display small and statistically insignificant coefficients, and Government services show no exporter-side effect. These cross-industry differences indicate that the benefits of digital infrastructure are concentrated in sectors where service provision and consumption can occur remotely.

Mobile network coverage produces a more mixed picture. Broader 2G-plus coverage is associated with higher overall trade flows, with exporter and importer elasticities of about 0.31 and 0.54, respectively, although the estimates for services are very large and imprecise, likely because 2G-plus coverage was near saturation in most countries. Higher-capacity 3G-plus coverage has no detectable effect on aggregate trade or on agricultural and manufacturing trade, but its elasticities are around 0.30 to 0.35 for services as a whole and are statistically significant for telecommunications and information services and education, and marginally so for finance. These findings suggest that both access and bandwidth are relevant: basic mobile networks expanded market participation on the extensive margin, while higher-capacity networks are associated with more trade in information-rich services, consistent with an intensive-margin channel.

In sum, the evidence indicates that digital connectivity has become an important determinant of international services trade. Internet usage and, to a lesser extent, mobile network coverage are associated with substantially more services trade, with the strongest effects in information-intensive, tradable service industries. The magnitudes observed imply that improvements in connectivity can deliver sizable trade gains, particularly in economies where digital infrastructure and regulatory quality evolve together. Policies that promote affordable internet access, expand high-speed mobile coverage, and ensure open and competitive service markets will be important for realizing the full potential of digital globalization.

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Figures and Tables

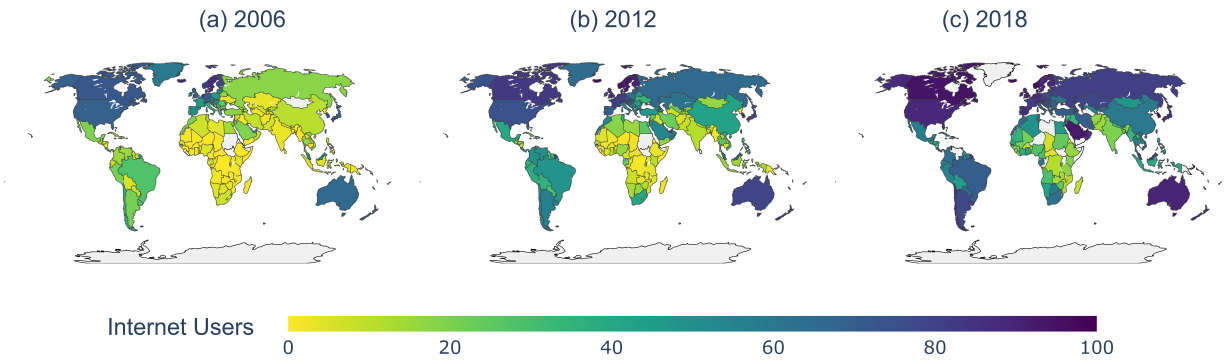


Figure 1: Internet User Percentage, 2006–2018

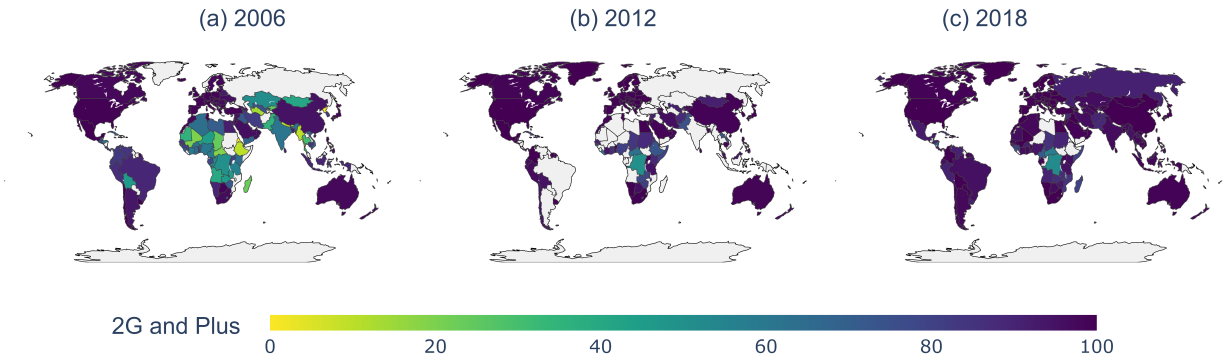


Figure 2: Mobile 2G+ Coverage Percentage, 2006–2018

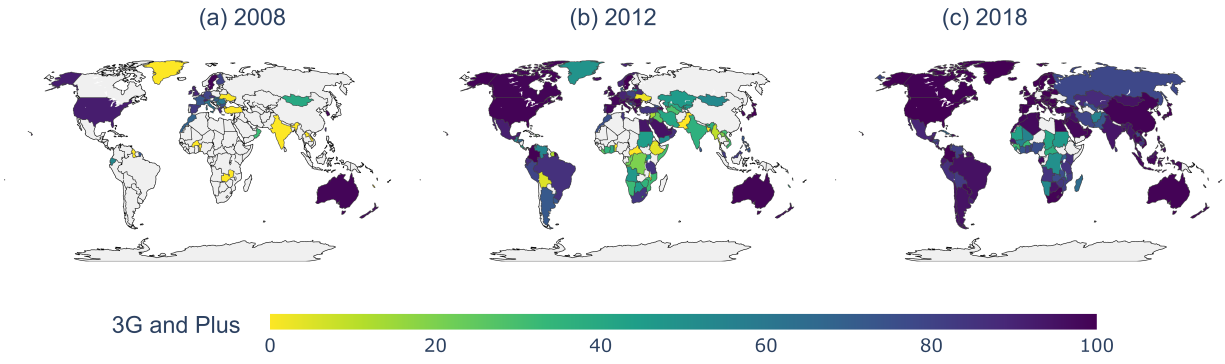


Figure 3: Mobile 3G+ Coverage Percentage, 2008–2018

Table 1: Effects of Internet Users

Panel A: First Stage	(1) Overall		(2) Agri	(3) Manuf	(4) Mining	(5) Services
FTA	0.443***		0.964***	0.569***	0.458***	0.329***
	(0.066)		(0.097)	(0.078)	(0.118)	(0.113)
ln(Distance)	−0.402***		−0.861***	−0.486***	−1.058***	−0.346***
	(0.043)		(0.074)	(0.054)	(0.089)	(0.052)
Contiguity	0.639***		0.548***	0.497***	0.419**	0.532***
	(0.087)		(0.161)	(0.097)	(0.206)	(0.118)
Common language	0.283***		0.198*	0.301***	0.454***	0.326***
	(0.056)		(0.109)	(0.069)	(0.131)	(0.091)
Colony ever	0.336***		0.022	0.209*	0.520**	0.666***
	(0.091)		(0.188)	(0.116)	(0.240)	(0.150)
Observations	1,497,078		444,685	633,334	332,095	86,651
R-squared	0.956		0.968	0.970	0.954	0.993
Panel B: Second Stage	**(1a) Overall**	**(1b) Overall**	**(2) Agri**	**(3) Manuf**	**(4) Mining**	**(5) Services**
Internet users_o	0.008**					
	(0.004)					
Internet users_d	0.013***					
	(0.004)					
$\ln(\text{Internet users})_o$		0.305***	0.111***	0.504***	0.104	0.713***
		(0.061)	(0.042)	(0.066)	(0.076)	(0.270)
$\ln(\text{Internet users})_d$		0.453***	0.320***	0.488***	0.718***	0.726***
		(0.052)	(0.053)	(0.068)	(0.059)	(0.266)
Observations	1,497,078	1,497,078	444,685	633,334	332,182	86,877
R-squared	0.671	0.673	0.787	0.875	0.807	0.677

Notes: Panel A presents the first-stage regressions. Panel B shows the second-stage results. Standard errors are clustered at the exporter–importer pair level and shown in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 2: Effects of 2G+ Mobile Internet Coverage, Second Stage

	(1a) Overall	(1b) Overall	(2) Agri	(3) Manuf	(4) Mining	(5) Services
2G+ coverage$_o$	0.006***					
	(0.002)					
2G+ coverage$_d$	0.010***					
	(0.003)					
ln(2G+ coverage)$_o$		0.314**	−0.089	0.522***	−0.032	3.777**
		(0.125)	(0.093)	(0.159)	(0.231)	(1.495)
ln(2G+ coverage)$_d$		0.535***	0.136	0.617***	1.117***	3.770**
		(0.150)	(0.147)	(0.163)	(0.312)	(1.487)
Observations	884,858	884,858	266,880	345,032	208,481	64,455
R-squared	0.681	0.680	0.804	0.873	0.811	0.686

Notes: Standard errors are clustered at the exporter–importer pair level and shown in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 3: Effects of 3G+ Mobile Internet Coverage, Second Stage

	(1a) Overall	(1b) Overall	(2) Agri	(3) Manuf	(4) Mining	(5) Services
3G+ coverage$_o$	−0.002					
	(0.002)					
3G+ coverage$_d$	−0.002					
	(0.002)					
ln(3G+ coverage)$_o$		0.008	−0.083	−0.049	0.016	0.318*
		(0.063)	(0.069)	(0.080)	(0.057)	(0.192)
ln(3G+ coverage)$_d$		0.023	−0.018	−0.081	0.179***	0.324*
		(0.062)	(0.052)	(0.078)	(0.064)	(0.193)
Observations	600,399	600,399	180,017	239,349	138,691	42,336
R-squared	0.661	0.661	0.800	0.859	0.809	0.696

Notes: Standard errors are clustered at the exporter–importer pair level and shown in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 4: Effects of Internet Users by Service Industry, Second Stage

	(1)	(2)	(3)	(4)	(5)
Panel A	154	155	156	157	158
$\ln(\text{internet users})_o$	0.504	1.057***	0.738***	0.424***	1.016***
	(0.761)	(0.279)	(0.261)	(0.138)	(0.329)
$\ln(\text{internet users})_d$	0.971	0.164	0.760***	0.762***	1.016***
	(0.767)	(0.151)	(0.256)	(0.167)	(0.328)
Observations	19,432	29,422	75,331	54,258	47,106
R-squared	0.672	0.857	0.565	0.814	0.521
	(1)	(2)	(3)	(4)	(5)
Panel B	159	160	161	162	163
$\ln(\text{internet users})_o$	0.814***	0.802***	0.420**	0.856***	0.729***
	(0.273)	(0.239)	(0.203)	(0.241)	(0.222)
$\ln(\text{internet users})_d$	0.809***	0.865***	0.414***	0.824***	0.666***
	(0.268)	(0.211)	(0.104)	(0.255)	(0.221)
Observations	54,244	56,911	51,032	68,077	70,523
R-squared	0.748	0.712	0.906	0.750	0.707
	(1)	(2)	(3)	(4)	(5)
Panel C	164	165	166	167	168
$\ln(\text{internet users})_o$	0.170	0.617***	0.809***	−0.020	−0.142
	(0.110)	(0.231)	(0.231)	(0.149)	(0.127)
$\ln(\text{internet users})_d$	0.046	0.518**	0.959***	0.572**	0.029
	(0.175)	(0.240)	(0.174)	(0.243)	(0.131)
Observations	13,292	31,206	40,321	47,620	31,984
R-squared	0.798	0.755	0.873	0.840	0.846
	(1)	(2)	(3)	(4)	(5)
Panel D	169	170			
$\ln(\text{internet users})_o$	0.774***	0.016			
	(0.290)	(0.104)			
$\ln(\text{internet users})_d$	0.759***	0.113			
	(0.292)	(0.130)			
Observations	45,056	13,152			
R-squared	0.498	0.687			

Notes: Industry IDs refer to the following service categories: 154 is manufacturing services on physical inputs owned by others; 155 is maintenance and repair services not included elsewhere (n.i.e.); 156 is transport; 157 is travel; 158 is construction; 159 is insurance and pension services; 160 is financial services; 161 is charges for the use of intellectual property n.i.e.; 162 is telecommunications, computer, and information services; 163 is other business services; 164 is heritage and recreational services; 165 is health services; 166 is education services; 167 is government goods and services n.i.e.; 168 is services not allocated; 169 is trade-related services; and 170 is other personal services. Standard errors are clustered at the exporter–importer pair level and shown in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 5: Effects of 2G+ Coverage by Service Industry, Second Stage

Panel A	(1) 154	(2) 155	(3) 156	(4) 157	(5) 158
$\ln(\text{2G+ coverage})_o$	−1.661	0.717	0.390	1.044*	5.797***
	(1.466)	(0.749)	(0.714)	(0.585)	(1.951)
$\ln(\text{2G+ coverage})_d$	1.860	0.336	1.353	2.390**	5.651***
	(1.690)	(0.607)	(1.598)	(1.061)	(1.928)
Observations	16,763	24,385	56,534	43,432	36,511
R-squared	0.690	0.864	0.571	0.817	0.519
Panel B	(1) 159	(2) 160	(3) 161	(4) 162	(5) 163
$\ln(\text{2G+ coverage})_o$	5.048**	3.709**	−0.207	3.823***	3.969**
	(2.005)	(1.829)	(0.438)	(1.289)	(1.619)
$\ln(\text{2G+ coverage})_d$	3.494	2.753	0.492	3.553**	3.333**
	(2.170)	(1.841)	(0.311)	(1.555)	(1.533)
Observations	41,598	43,949	39,275	52,737	54,236
R-squared	0.752	0.711	0.905	0.750	0.711
Panel C	(1) 164	(2) 165	(3) 166	(4) 167	(5) 168
$\ln(\text{2G+ coverage})_o$	−1.247	5.419***	5.778***	0.521	0.098
	(0.000)	(1.841)	(1.964)	(0.493)	(0.186)
$\ln(\text{2G+ coverage})_d$	−0.255	5.478***	2.335**	−0.124	0.190
	(0.000)	(1.789)	(1.182)	(0.207)	(0.144)
Observations	11,482	25,800	33,027	37,035	24,370
R-squared	0.791	0.768	0.882	0.843	0.863
Panel D	(1) 169	(2) 170	(3)	(4)	(5)
$\ln(\text{2G+ coverage})_o$	3.779***	0.674			
	(1.370)	(0.000)			
$\ln(\text{2G+ coverage})_d$	3.610**	0.958			
	(1.462)	(0.000)			
Observations	36,514	11,497			
R-squared	0.514	0.682			

Notes: Industry IDs refer to the following service categories: 154 is manufacturing services on physical inputs owned by others; 155 is maintenance and repair services not included elsewhere (n.i.e.); 156 is transport; 157 is travel; 158 is construction; 159 is insurance and pension services; 160 is financial services; 161 is charges for the use of intellectual property n.i.e.; 162 is telecommunications, computer, and information services; 163 is other business services; 164 is heritage and recreational services; 165 is health services; 166 is education services; 167 is government goods and services n.i.e.; 168 is services not allocated; 169 is trade-related services; and 170 is other personal services. Standard errors are clustered at the exporter–importer pair level and shown in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 6: Effects of 3G+ Coverage by Service Industry, Second Stage

	(1)	(2)	(3)	(4)	(5)
Panel A	154	155	156	157	158
$\ln(\text{3G+ coverage})_o$	0.049	0.255	0.312*	0.168	0.458**
	(0.212)	(0.166)	(0.178)	(0.130)	(0.198)
$\ln(\text{3G+ coverage})_d$	−0.143	0.129***	0.311*	0.262**	0.460**
	(0.223)	(0.047)	(0.174)	(0.117)	(0.199)
Observations	12,944	18,480	36,589	28,046	23,583
R-squared	0.722	0.873	0.577	0.836	0.533
	(1)	(2)	(3)	(4)	(5)
Panel B	159	160	161	162	163
$\ln(\text{3G+ coverage})_o$	0.364*	0.342*	0.205	0.356**	0.312**
	(0.207)	(0.199)	(0.152)	(0.158)	(0.153)
$\ln(\text{3G+ coverage})_d$	0.366*	0.342*	0.103***	0.395**	0.301*
	(0.206)	(0.189)	(0.035)	(0.188)	(0.170)
Observations	27,082	28,247	25,326	34,298	35,221
R-squared	0.766	0.726	0.909	0.779	0.729
	(1)	(2)	(3)	(4)	(5)
Panel C	164	165	166	167	168
$\ln(\text{3G+ coverage})_o$	−0.287	0.298	0.385**	0.158***	−0.642
	(0.326)	(0.182)	(0.189)	(0.059)	(0.610)
$\ln(\text{3G+ coverage})_d$	−0.332	0.300*	0.282***	0.039	0.081
	(0.357)	(0.180)	(0.086)	(0.075)	(0.208)
Observations	7,971	17,131	21,138	24,077	8,298
R-squared	0.800	0.804	0.892	0.854	0.795
	(1)	(2)	(3)	(4)	(5)
Panel D	169	170			
$\ln(\text{3G+ coverage})_o$	0.300	−0.065			
	(0.193)	(0.130)			
$\ln(\text{3G+ coverage})_d$	0.300	−0.033			
	(0.193)	(0.120)			
Observations	23,432	7,720			
R-squared	0.589	0.715			

Notes: Industry IDs refer to the following service categories: 154 is manufacturing services on physical inputs owned by others; 155 is maintenance and repair services not included elsewhere (n.i.e.); 156 is transport; 157 is travel; 158 is construction; 159 is insurance and pension services; 160 is financial services; 161 is charges for the use of intellectual property n.i.e.; 162 is telecommunications, computer, and information services; 163 is other business services; 164 is heritage and recreational services; 165 is health services; 166 is education services; 167 is government goods and services n.i.e.; 168 is services not allocated; 169 is trade-related services; and 170 is other personal services. Standard errors are clustered at the exporter–importer pair level and shown in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

A Appendix: Additional Figures and Tables

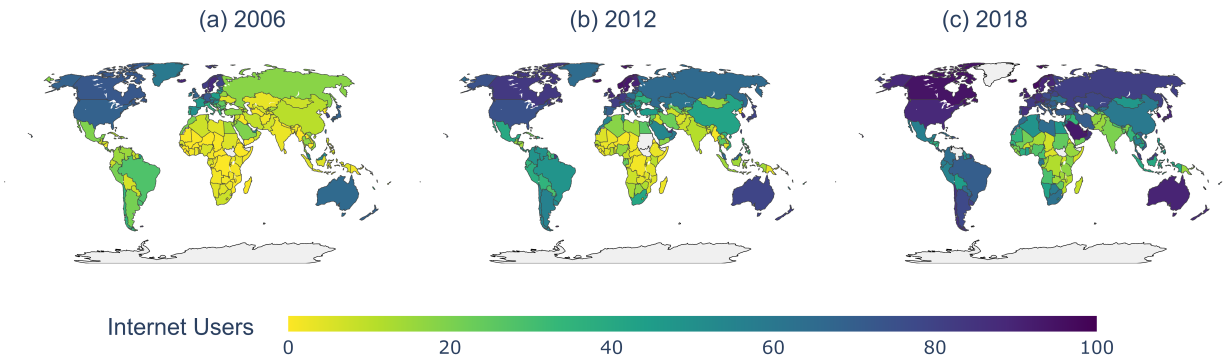


Figure A.1: Internet User Percentage (2-year Interpolation), 2006–2018

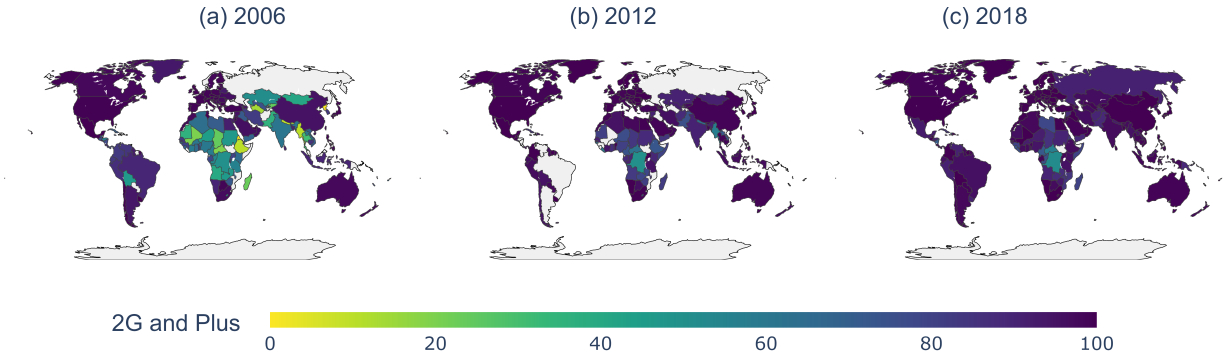


Figure A.2: Mobile 2G+ Coverage Percentage (2-year Interpolation), 2006–2018

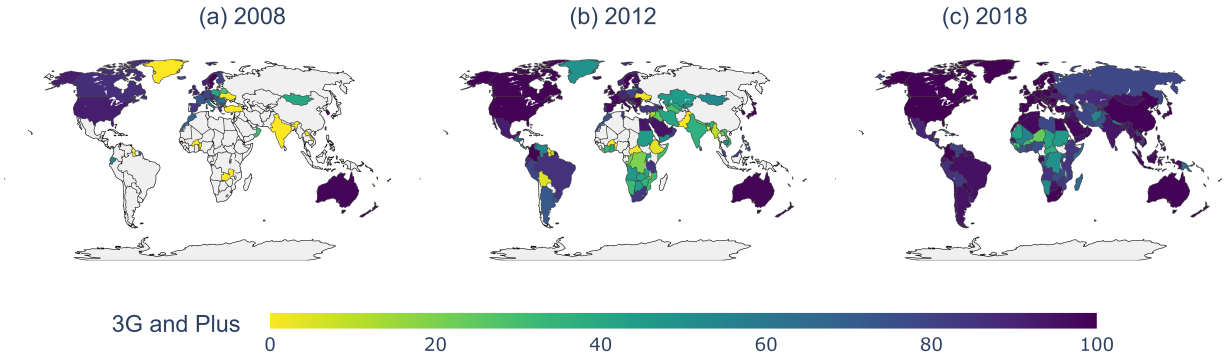


Figure A.3: Mobile 3G+ Coverage Percentage (2-year Interpolation), 2008–2018

Table A.1: Effects of 2G+ Mobile Internet Coverage, First Stage

	(1a) Overall	(1b) Overall	(2) Agri	(3) Manuf	(4) Mining	(5) Services
FTA	0.370***	0.370***	0.996***	0.481***	0.547***	0.262**
	(0.069)	(0.069)	(0.094)	(0.076)	(0.115)	(0.126)
ln(Distance)	−0.387***	−0.387***	−0.867***	−0.485***	−1.032***	−0.350***
	(0.044)	(0.044)	(0.079)	(0.055)	(0.093)	(0.055)
Contiguity	0.716***	0.716***	0.678***	0.560***	0.617***	0.594***
	(0.086)	(0.086)	(0.162)	(0.090)	(0.184)	(0.124)
Common language	0.250***	0.250***	0.190*	0.269***	0.460***	0.317***
	(0.057)	(0.057)	(0.105)	(0.068)	(0.129)	(0.094)
Colony ever	0.334***	0.334***	0.080	0.167	0.548**	0.692***
	(0.095)	(0.095)	(0.195)	(0.124)	(0.255)	(0.156)
Observations	938,063	938,063	280,731	372,826	218,398	65,112
R-squared	0.959	0.959	0.971	0.972	0.964	0.993

Notes: Standard errors are clustered at the exporter–importer pair level and shown in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table A.2: Effects of 3G+ Mobile Internet Coverage, First Stage

	(1a) Overall	(1b) Overall	(2) Agri	(3) Manuf	(4) Mining	(5) Services
FTA	0.427***	0.427***	0.874***	0.542***	0.538***	0.360***
	(0.064)	(0.064)	(0.095)	(0.076)	(0.100)	(0.100)
ln(Distance)	−0.343***	−0.343***	−0.866***	−0.455***	−1.009***	−0.335***
	(0.044)	(0.044)	(0.073)	(0.057)	(0.097)	(0.050)
Contiguity	0.757***	0.757***	0.559***	0.615***	0.542***	0.528***
	(0.088)	(0.088)	(0.148)	(0.094)	(0.183)	(0.117)
Common language	0.299***	0.299***	0.203**	0.247***	0.472***	0.441***
	(0.056)	(0.056)	(0.100)	(0.070)	(0.122)	(0.091)
Colony ever	0.338***	0.338***	0.069	0.272**	0.391	0.708***
	(0.097)	(0.097)	(0.198)	(0.116)	(0.250)	(0.157)
Observations	641,644	641,644	189,215	263,829	145,614	42,644
R-squared	0.957	0.957	0.968	0.963	0.956	0.994

Notes: Standard errors are clustered at the exporter–importer pair level and shown in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.